67. At all times, an urn contains $N$ balls—some white balls and some black balls. At each stage, a coin having probability $p, 0 < p < 1$, of landing heads is flipped. If heads appears, then a ball is chosen at random from the urn and is replaced by a white ball; if tails appears, then a ball is chosen from the urn and is replaced by a black ball. Let $X_n$ denote the number of white balls in the urn after the $n$th stage.

    (a) Is $\{X_n, n \geq 0\}$ a Markov chain? If so, explain why.

    (b) What are its classes? What are their periods? Are they transient or recurrent?

    (c) Compute the transition probabilities $P_{ij}$.

    (d) Let $N = 2$. Find the proportion of time in each state.

    (e) Based on your answer in part (d) and your intuition, guess the answer for the limiting probability in the general case.

    (f) Prove your guess in part (e) either by showing that Theorem (4.1) is satisfied or by using the results of Example 4.37.

    (g) If $p = 1$, what is the expected time until there are only white balls in the urn if initially there are $i$ white and $N - i$ black?

68. (a) Show that the limiting probabilities of the reversed Markov chain are the same as for the forward chain by showing that they satisfy the equations

$$\pi_j = \sum_i \pi_i Q_{ij}$$

70. A total of $m$ white and $m$ black balls are distributed among two urns, with each urn containing $m$ balls. At each stage, one ball is drawn uniformly at random from each urn and the two drawn balls are interchanged. Let $X_n$ denote the number of black balls in urn 1 after the $n$-th interchange.

    (a) Give the transition probabilities of the Markov chain $\{X_n, n \geq 0\}$.

    (b) Without computations, what are the limiting probabilities?

    (c) Find the limiting probabilities and show that the chain is time reversible.